

Exact partition function and target probability for generalized BITCOPY in 1D SDC

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1. Introduction

Theorem 17 and Corollary 18 of J. Petrack, C. G. Evans, A. Cervera Roldan, M. Enayati, and D. Woods [1] give an exact value for the equilibrium target probability of a BITCOPY SDC system [2] of length N , and show that a compute domain energy that scales as $\Delta G_c^\circ/RT = -\ln N + O(1)$ is sufficient to maintain a constant target probability with increasing N , but both are subject to the assumption that all scaffold sites are filled, and partially-filled configurations can be ignored. That assumption is not necessary: both results can be obtained for the general case that includes partially-filled configurations, using an approach adapted from exact solutions for 1D Ising models [3].

Here, we calculate an exact, closed-form partition function for a k -copy system. While directly a slight generalization of the BITCOPY system ($k = 2$), some other systems are effectively the same for the purpose of state enumeration: a simple relabelling of tiles at each position from incrementing to copying converts a COUNT modulo k to k -copy, for example. Since the target state is known, and its energy easily calculated, the partition function allows us to obtain an exact, closed form of the target state probability.

While the resulting partition function and target state probability are complicated and, compared to the all-filled simplification, difficult to intuitively understand, asymptotically the same scaling of ΔG_c° , of $-O(\ln N)$, suffices to maintain a fixed equilibrium target probability: intuitively, the increased compute domain strength necessary to maintain the target's favorability over other all-filled configurations also suffices to ensure sufficient unfavorability of partially-filled configurations.

The transfer matrix form used here can be used for any 1D SDC program, though the form is less amenable to computational implementation than algorithmic approaches [4]; it is better suited to circumstances where properties of the program may allow for simplifications, as occurs with k -copy. The almost direct applicability of an Ising model technique also illustrates the close links between Ising models and SDC: in some sense, abstracted away from DNA and scaffolds, SDC can be seen as a generalization of the Ising model. On those lines, while considerably more difficult, approaches conceptually similar to that used for the 1D Ising model can also be applied to the 2D Ising model [3], suggesting that a careful review of Ising model literature could prove fruitful in studying more complex SDC systems.

2. Exact partition function

Theorem 1. For a k -copy system ($k \geq 2$) of length N in the bind/unbind model, where every tile has the same concentration c , every compute domain has the same energy ΔG_c° , and every scaffold domain

has the same energy ΔG_p° , the equilibrium probability of the target state is $p(\text{target}) = Z^{-1}a^N b^{N-1}$, where $a = e^{-(\Delta G_p^\circ - \mu)/RT}$, $b = e^{-\Delta G_c^\circ/RT}$, $m = k - 1$, $\mu = RT \ln(cu_0^{-1})$,

$$Z = \frac{(\lambda_+^{N-1} - \lambda_-^{N-1})(ma^2 + a^2b + (m+2)a + 1)}{\lambda_+ - \lambda_-} - \frac{(\lambda_- \lambda_+^{N-1} - \lambda_+ \lambda_-^{N-1})(a+1)}{\lambda_+ - \lambda_-}$$

and

$$\lambda_\pm = \frac{1 + am + ab \pm \sqrt{(1 + am + ab)^2 - 4(ab - a)}}{2}$$

Proof. We start with the usual configuration energy for 1D SDC:

$$G(\alpha) = \sum_{i=1}^N [\Delta G_p^\circ(i) - \mu(i)] + \sum_{i=1}^{N-1} \Delta G_c^\circ(i, i+1) \quad ,$$

where any term involving an unoccupied site = 0, and otherwise $\Delta G_p^\circ(i) = \Delta G_p^\circ$, $\Delta G_c^\circ(i, i+1) = \Delta G_c^\circ$ if the tile at i and $i+1$ represent the same value (and is = 0 otherwise), and $\mu(i) = RT \ln(cu_0^{-1})$ (with $u_0 = 1$ M).

We also have the usual $p(\alpha) = Z^{-1}e^{-G(\alpha)/RT}$, where

$$Z = \sum_{\alpha} e^{-G(\alpha)/RT} \quad .$$

The form of Z means that it can be seen as summing different possible configurations, while multiplying the (exponentiated) contributions of each position to G . The $G(\alpha)$ in 1D SDC can also be seen as having the contribution of each position be determined only by the tile at that position, and at the previous position. This allows a 'transfer matrix' approach: [3, Ch. 2] Z can be recast as a product of many previous-possible-states $\times$ current-possible-states matrices, where each location represents the combination of one possible previous position state and one possible current position state.

For a k -copying program, the first position is a non-competitive anchor, and all subsequent positions have k possible attachments. For a configuration α , the first position has only two possible states: filled or empty. Each subsequent position then contributes to the energy based on both the tile at that position, and at the previous position.

For convenience, let $a = e^{-(\Delta G_p^\circ - \mu)/RT}$, $b = e^{-\Delta G_c^\circ/RT}$, and $m = k - 1$. Then consider positions from left to right:

- For the anchor, the position can either be full (a) or empty (1).
- At each subsequent position, if the previous position was full, then the current position can be full but mismatch in m different ways (ma), full and match (ab), or empty (1). If the previous position was empty, then the current position can either be full in one of $m+1$ states with no compute domain bond (a for each), or empty (1).

- Once finished, we simply need to sum the different paths.

This gives us the product of matrices

$$Z = (a \ 1) \begin{pmatrix} ma + ab & 1 \\ (m+1)a & 1 \end{pmatrix}^{N-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

While it might intuitively make sense to use a $k+1 \times k+1$ matrix, with entries for empty and k possible filled (one matched, $m = k-1$ mismatched) subsequent positions, since the same possible transitions exist regardless of what value was in the previous position, we can combine terms to use a simpler 2×2 matrix. We can then use Sylvester's formula for a function of a diagonalizable matrix, which is

$$f(A) = \sum_{i=1}^d f(\lambda_i) \prod_{j=1, j \neq i}^d \frac{1}{\lambda_i - \lambda_j} (A - \lambda_j I),$$

where λ_i are the d distinct eigenvalues of A .

In our simple case, we have

$$M^{N-1} = \frac{(\lambda_+^{N-1} - \lambda_-^{N-1})M - (\lambda_- \lambda_+^{N-1} - \lambda_+ \lambda_-^{N-1})I}{\lambda_+ - \lambda_-}$$

where $\lambda_{\pm}$ are the two eigenvalues of M that can be simply derived from $\det(M - \lambda_{\pm}I) = 0$, but take the unwieldy quadratic formula form

$$\begin{aligned} \lambda_{\pm} &= \frac{1 + am + ab \pm \sqrt{(1 + am + ab)^2 - 4(ab - a)}}{2} \\ &= \frac{1 + a(m+b) \pm \sqrt{(a(m+b) - 1)^2 + 4a(m+1)}}{2}. \end{aligned}$$

which, for $m \geq 0, a > 0, b > 0$, is safely non-degenerate. Returning to Z , we get

$$\begin{aligned} Z &= \frac{(\lambda_+^{N-1} - \lambda_-^{N-1})(ma^2 + a^2b + (m+2)a + 1)}{\lambda_+ - \lambda_-} \\ &\quad - \frac{(\lambda_- \lambda_+^{N-1} - \lambda_+ \lambda_-^{N-1})(a+1)}{\lambda_+ - \lambda_-}. \end{aligned}$$

and since the target state Boltzmann weight can be simply expressed as $e^{-G(\text{target})/RT} = a^N b^{N-1}$, (intuitively, N tiles and $N-1$ compute domain bonds), we have

$$p(\text{target}) = Z^{-1} a^N b^{N-1}. \quad \square$$

3. Asymptotic behavior

Corollary 2. For fixed unitless scaffold energy $\Delta G_p^\circ/RT$, concentration, and k , the $\Delta G_c^\circ/RT$ required to maintain a fixed target state probability $P \in (0, 1)$ at equilibrium for a k -copy system as $N \rightarrow \infty$ satisfies $\Delta G_c^\circ/RT = \ln(-\ln P) - \ln(k-1) - \ln N + o(1)$.

Proof. As increasing compute domain strength with N is required for keeping a fixed $p(\text{target} \mid \text{filled})$ (per [1]), we will begin by guessing a similar expression of $b = qN$ for some $q > 0$ (recall $b \equiv \exp(-\Delta G_c^\circ/RT)$). We need to obtain asymptotic expressions for $\lambda_{\pm}$ in this regime. We start by considering the discriminant in isolation, that is, by having

$$\lambda_{\pm} = \frac{1}{2}(1 + am + ab \pm \sqrt{D})$$

where

$$D = (1 + am + ab)^2 - 4(ab - a).$$

Then, applying our expression $b = qN$, distributing, and keeping only terms with at least a factor of N , we have

$$\begin{aligned} D &= (aqN)^2 + 2(ma - 1)aqN + O(1) \\ \sqrt{D} &= aqN \sqrt{1 + 2(ma - 1)(aqN)^{-1} + O((aqN)^{-2})} \\ \sqrt{D} &\approx aqN + (ma - 1). \end{aligned}$$

We then can return to

$$\begin{aligned} \lambda_+ &= aqN + ma + O(N^{-1}) \\ \lambda_- &= 1 + O(N^{-1}), \end{aligned}$$

and move to Z , keeping only terms involving N :

$$\begin{aligned} Z &\approx \frac{\lambda_+^{N-1} a^2 b}{\lambda_+ - \lambda_-} \\ &\approx \lambda_+^{N-2} a^2 qN \\ &\approx a^N (qN)^{N-1} \left(1 + \frac{m}{qN}\right)^{N-2} \end{aligned}$$

and thus

$$\begin{aligned} p(\text{target}) &\approx \left(1 + \frac{m}{qN}\right)^{2-N} \\ \ln p &\approx (2 - N) \ln \left(1 + \frac{m}{qN}\right) \\ \ln p &\approx -N \frac{m}{qN} + O(N^{-1}) = -\frac{m}{q} + O(N^{-1}) \\ q &\approx -\frac{m}{\ln p} \end{aligned}$$

We can formalize this scaling by considering $p(\text{target}, b, N)$ with fixed m and a . For any relevant N ($N \geq 2$ is necessary to have any compute domain bonds at all), an increase in compute domain strength (larger positive b) must increase the Boltzmann weight of the target configuration more than any other configuration: for deterministic systems, the target configuration has more compute domain bonds than any other configuration, and thus a larger power of b in its Boltzmann weight. Thus, the target state's share of the partition function must increase, and $p(\text{target}, b, N)$ is continuous and strictly increasing

in b . Also, $\lim_{b \rightarrow 0} p(\text{target}, b, N) = 0$ (infinitely unfavorable compute domains, not simply weak) and $\lim_{b \rightarrow \infty} p(\text{target}, b, N) = 1$ (infinitely strong compute domains). There is thus a unique $b(N, P)$ such that $p(\text{target}, b(N, P), N) = P$ for some desired $0 < P < 1$.

We can define $q_0 = -m/(\ln P)$. For $q > 0$, $\lim_{N \rightarrow \infty} p(\text{target}, qN, N) = e^{-m/q}$, so for any other choices $0 < q_- < q_0 < q_+$, and for sufficiently large N , $p(\text{target}, q_- N, N) < P < p(\text{target}, q_+ N, N)$, and strict monotonicity gives $q_- N < b(N, P) < q_+ N$. As this must hold for q_- and q_+ arbitrarily close to q_0 , $\lim_{N \rightarrow \infty} \frac{b(N, P)}{N} = \frac{m}{-\ln P}$, and $b(N, P) = q_0 N + o(N)$. Thus

$$\begin{aligned} \frac{\Delta G_c^\circ}{RT} &= -\ln b(N, P) \\ &= \ln(-\ln P) - \ln m - \ln N + o(1) \quad . \quad \square \end{aligned}$$

References

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